

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 4: Moisture, Humidity & Atmospheric Stability

Syllabus Source: CAE Oxford (pp. 99-136) | Pass Target: >= 90% AviationExam

1. HUMIDITY CONCEPTS & THE DEW POINT

Water vapor is the driving fuel of atmospheric thermodynamics:

- Relative Humidity (RH): Ratio of actual water vapor pressure to the saturation vapor pressure at that temperature, expressed as %: $RH = (Actual\ Vapor\ Pressure / Saturation\ Vapor\ Pressure) \times 100$.
- Dew Point (DP / Td): The temperature to which unsaturated air must be cooled at constant pressure to become fully saturated (RH = 100%). Small temperature-dew point spread indicates imminent fog or cloud formation!

2. ADIABATIC LAPSE RATES & STABILITY

As an air parcel rises, it expands due to decreasing external pressure and cools without exchanging heat with surrounding air (adiabatic process):

- Dry Adiabatic Lapse Rate (DALR): Applies to unsaturated air: Constant at 3.0°C / 1,000 ft (1.0°C / 100 m = 9.8°C / km).
- Saturated Adiabatic Lapse Rate (SALR): Applies once condensation occurs (saturated air): Variable, approximately 1.5°C / 1,000 ft (0.5°C to 0.7°C / 100 m). SALR is lower than DALR because latent heat of condensation (600 cal/g) is released into the parcel, retarding its cooling!
- Environmental Lapse Rate (ELR): The actual measured temperature profile of the ambient atmosphere.

Atmospheric Stability State	Mathematical Criterion (ELR vs DALR / SALR)	Associated Weather & Cloud Types
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Absolute Stability	ELR < SALR (Lapse rate is shallow or inverted, e.g. < 1.5°C/1,000 ft).	No vertical convection. Stratiform flat clouds (Stratus), smooth air, poor surface visibility/fog.
Absolute Instability	ELR > DALR (Lapse rate is steep, > 3.0°C/1,000 ft / superadiabatic).	Violent convective up/downdrafts, Cumulonimbus (CB), severe turbulence, good visibility outside showers.
Conditional Instability	SALR < ELR < DALR (ELR is between 1.5°C and 3.0°C/1,000 ft).	Stable if air is dry; violently UNSTABLE if lifted to its saturation level (LCL)!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Convective Cloud Base (LCL) Est

Problem: Surface Temperature = +26 C, Surface Dewpoint = +16 C. Step 1: Calculate Dewpoint Depression (Spread) = 26 C - 16 C = 10 C. Step 2: Combined cooling rate of ascending parcel vs dewpoint = 2.5 C per 1,000 ft (or 400 ft per 1 C). Step 3: Cloud Base (ft AGL) = (Spread / 2.5) x 1,000 = (10 / 2.5) x 1,000 = 4,000 ft AGL. Fast Rule: Base = Spread x 400 ft = 10 x 400 = 4,000 ft AGL!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)**[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Convective Cloud Base Height**

SCENARIO (Extremely Common EASA Exam Question): Surface weather observations report: - Surface Temperature = +22°C - Surface Dew Point = +12°C - Airfield elevation = 500 ft MSL QUESTION: At what altitude MSL will the base of convective Cumulus clouds form? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Temperature - Dew Point Spread at surface: - Spread = Surface Temperature - Dew Point - Spread = 22°C - 12°C = 10°C. Step 2: Understand how the Spread closes with altitude in a rising parcel: - Rising dry parcel cools at DALR = 3.0°C per 1,000 ft. - Dew point also drops slightly as pressure falls = 0.5°C per 1,000 ft. - Net closing rate of the spread = 3.0°C - 0.5°C = 2.5°C per 1,000 ft (or 400 ft per °C)! Step 3: Calculate Cloud Base Height Above Ground Level (AGL): - Cloud Base AGL = (Spread / 2.5°C) x 1,000 ft - Cloud Base AGL = (10°C / 2.5°C) x 1,000 ft = 4 x 1,000 ft = 4,000 ft AGL! Step 4: Convert Height AGL to Altitude MSL (Add airfield elevation): - Cloud Base MSL = Height AGL + Field Elevation - Cloud Base MSL = 4,000 ft + 500 ft = 4,500 ft MSL! FINAL ANSWER: Cloud base forms at 4,000 ft AGL (4,500 ft MSL).

[!] EXAM TRAP (AVIATIONEXAM) - AGL vs MSL Cloud Base Trap

- The formula $(T - DP)/2.5$ gives cloud base ABOVE GROUND LEVEL (AGL)! - If the exam question asks for ALTITUDE MSL, you MUST ADD aerodrome elevation! - Forgetting to add elevation is the most frequent trap in this question.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Why is the Saturated Adiabatic Lapse Rate (SALR) lower than the Dry Adiabatic Lapse Rate (DALR)?**

- [A]: Saturated air is heavier
- [B]: Because LATENT HEAT of condensation is released when water vapor condenses, warming the rising parcel
- [C]: Friction with falling rain
- [D]: Sun heating

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. As water vapor condenses into water droplets, latent heat (approx 2.5 x 10⁶ J/kg) is released into the air parcel, partially offsetting adiabatic expansion cooling.

Q2: If the ambient Environmental Lapse Rate (ELR) is 2.2°C per 1,000 ft, what is the stability state of the atmosphere?

- [A]: Absolute stability
- [B]: Absolute instability
- [C]: CONDITIONAL INSTABILITY (because ELR lies between SALR ~1.5°C and DALR 3.0°C)

- [D]: Neutral

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. When ELR is between SALR and DALR, the atmosphere is conditionally unstable: stable if unsaturated, but unstable if forced to saturation.

Q3: What type of cloud is produced when an absolutely STABLE layer of air is forced to rise over a mountain r

- [A]: Cumulonimbus with severe hail
- [B]: STRATIFORM clouds (smooth sheets such as Stratus or Altostratus) without vertical development
- [C]: Isolated towering cumulus
- [D]: Mammatus

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In stable air, displaced air resists vertical motion, producing flat, horizontal layer clouds without thermal turbulence.